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Information and Network Security

CA-3: Case Study Report

Titled

"NCBA: Non-Person Entity Certificate Behavior Analytics Framework"

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NCBA: Non-Person Entity Certificate Behavior Analytics Framework

Abstract

Machine identities are everywhere now like microservices, containers, IoT devices, APIs; and they're multiplying fast. Recent numbers show machine identities outnumber human ones 43 to 1 [3], [5] in most enterprise environments. That's a huge shift from just a few years ago, but security frameworks haven't really caught up. Most still focus on human users while treating machine identities as an afterthought.

The problem is that these non-person entities (NPEs) usually get long-lived certificates, valid for months or even years [6], [7]. When one gets compromised, attackers have plenty of time to move laterally across the network before anyone notices. This report proposes NCBA, a framework that tries to address this by combining short-lived certificates (under 24 hours), machine learning-based behavioral analytics, and blockchain audit trails [7], [9], [12]. The idea is to detect compromised certificates quickly and respond automatically, even in environments where you can't install endpoint agents like legacy industrial systems or cheap IoT sensors.

Keywords:

Non-Person Entity, Machine Identity Management, Certificate Lifecycle Automation, Behavioral Analytics, Zero Trust Architecture, Blockchain Transparency, Lateral Movement Detection

1. Introduction

Modern digital systems are no longer driven primarily by human users. Instead, they are powered by an ever-growing ecosystem of machine identities like microservices communicating within cloud environments, APIs enabling integrations, containers scaling dynamically, and IoT devices operating at the edge. These non-person entities (NPEs) now form the backbone of enterprise infrastructure, often outnumbering human identities by a significant margin [3], [5]. Despite this shift, cybersecurity practices have not evolved at the same pace. Most existing identity and access management frameworks remain centered around human authentication patterns, leaving machine identities insufficiently monitored and protected.

In our research, we found that one of the most critical risks associated with NPEs lies in their reliance on digital certificates for authentication. These certificates are typically long-lived [6], remaining valid for months or even years. While this reduces operational complexity, it also creates a substantial security gap. If a certificate is

compromised, attackers can exploit it to impersonate legitimate services and move laterally across networks without detection [3]. Unlike human accounts, which may trigger alerts through unusual login behavior, machine identities often operate silently and predictably, making misuse harder to identify using traditional security tools.

Existing solutions primarily address certificate lifecycle management which focuses on issuance, renewal, and revocation but lack visibility into how certificates are actually used during their lifetime. Similarly, behavioral analytics systems [8][11] such as User and Entity Behavior Analytics (UEBA) are designed around human activity patterns and fail to effectively model machine behavior, which tends to be structured, repetitive, and service-oriented. This gap becomes even more pronounced in environments where endpoint security agents cannot be deployed, such as legacy industrial systems or resource-constrained IoT devices [5].

To address these challenges, our project introduces the Non-Person Entity Certificate Behavior Analytics (NCBA) Framework, a comprehensive approach that combines short-lived certificates, machine-specific behavioral analytics, and tamper-proof audit mechanisms. NCBA aims to detect and contain certificate misuse in near real-time by reducing certificate validity to hours instead of months[7], continuously analyzing network-level behavior, and enabling automated response mechanisms. This framework is designed with practicality in mind, particularly for environments where traditional endpoint-based security solutions are not feasible like legacy systems.

Through this approach, NCBA shifts the focus from static identity validation to continuous trust evaluation, aligning with Zero Trust principles and modern enterprise needs [3]. It ensures that in today’s distributed and automated systems, securing machine identities is not just an extension of human-centric security but a fundamentally different challenge that requires dedicated solutions.

2. Problem Statement

The proliferation of non-person entities (NPEs) like APIs, microservices, containers, and IoT devices is creating several problems in today's cybersecurity landscape. Machine identities surpass human identities [3][5] but current cybersecurity measures are still geared towards people. Most of the time, these machine identities have lifetime certificates that are not revoked after a certain period of validity. If hackers get their hands on these credentials, they will be able to masquerade themselves as the legitimate identity and traverse through your network infrastructure undetected [6][7].

Most of the current technologies deal only with the certificate's lifespan. They handle issuing and renewing certificates but do not track how the certificate is being used throughout its lifespan. Traditional behavior analysis systems are built with human behaviors in mind and do not work in scenarios where the entity under observation is a machine [8][11]. Therefore, there is no way to detect any suspicious behavior involving machines. Another problem that arises with machines and their corresponding certificates is that some legacy or low-resource systems are unable to accommodate an endpoint security agent.

3. Objectives

Mainly, this project aims at developing a security architecture that helps in identifying and mitigating any form of certificate misuse. This can be achieved by minimizing the risk posed by machine identities through the use of short certificate lifecycles, behavior-based analytics targeting machine activities, and monitoring on the network level. Moreover, there will be an emphasis on automation of actions like certificate revocation and blocking of the network to mitigate any form of attacks and minimize response time.

4. Literature Review

Below we have broken down our findings on the current implementations and research around this topic in some key categories, that cumulatively shape our project. We encountered many academic papers published within the last five years, highlighting the rising trends in Identity Management and Non-Person Entities (NPEs).

A. Machine Identity Management

Machine identities (Non-Person Entities) comprise microservices, containers, APIs, and Internet of Things (IoT) devices. The management of these identities has evolved significantly over the years, especially due to the increasing number of machine identities as compared to that of humans.

To efficiently manage the workload identity, solutions like *Secure Production Identity Framework for Everyone (SPIFFE)* [1][5] have become the standard for identifying and securing communications between workloads though short-lived X.509 certificates or JWT tokens called SVIDs (SPIFFE Verifiable Identity Documents).

According to a research done by *“Establishing Workload Identity for Zero Trust CI/CD: From Secrets to SPIFFE-Based Authentication,”* by *Garg, Patel and Sharma in*

2025, 67% of credential-related incidents are reduced in the production environments when static secrets are shifted to SPIFEE-based identities [1].

However, the drawbacks of this standard include:

- As evident in another research titled "*Credential Broker Patterns for Secure CI/CD" by Chauhan, Verma and Singh (2025)* [4]*,* even though credential broker patterns enable short-lived, policy-driven credentials across trust domains, current implementations still **lack the integration of behavioral monitoring**.
- The model mainly emphasizes authentication and identity issuance in service-to-service communications with **no provision for behavioral monitoring or anomaly detection.**
- No mechanism of continuously monitoring these identities or their abuse.

B. Certificate Lifecycle Automation

Certificate Lifecycle Management (CLM) has become a crucial focus area as organizations are struggling to manage the growing volume of machine identities [2][6]. CLM Tools automate:

- Issue of Certificates
- Renewal
- Revocation

Some common commercial CLM tools include Venafi, Keyfactor, and HashiCorp Vault, which prevent downtime due to expired certificates.

Research such as *"Identity Control Plane: The Unifying Layer for Zero Trust Infrastructure," by Patil, Kumar, and Joshi (2025)* state that CLM systems have evolved to reduce manual workload by about 90% [2]. This automation improves availability and compliance but lacks focus on detecting compromise, as explicitly noted in the paper.

As seen through another study where *Rodriguez and Martinez (2024)* conducted an extensive analysis of CLM tools, titled *"Automated Certificate Management at Scale: A 5-Year Industry Analysis",* which included evaluating 150 organizations' deployments of the tools. Their findings show that while the number of outages related to certificates has decreased by 78%, there has been a 34% increase in the number of security incidents related to compromised certificates [6]. The authors attribute this increase to what they call the "automation gap": where organizations automate the issuance and renewal process but do not monitor or respond to issues.

On the other hand, *Williams and Thompson (2023)* present additional research on *"Short-Lived Certificates: Security Benefits and Operational Challenges*" through ACM Digital Library. Their research determines that by shortening the certificate lifetime from 12-months to 24-hours, the organization only has a 1.4% chance of being exposed to an attack [7].

Therefore the major gap in CLM currently is that:

- Most existing certificate automation frameworks operate as standalone systems that do not contain real-time monitoring nor anomaly detection for the use of certificates.

C. Behavioral Analysis

According to the growing technological standards, approximately 50% of large enterprises are expected to implement advanced behavioral analytics in Identity Management Systems by 2028 [8]. User and Entity Behavior Analysis (UEBA) is a technology widely applied in cybersecurity for this use case and identifying anomalies.

- The technology is based on pattern recognition such as login activity, usage frequency, and other behavioral patterns.
- Popular algorithms include Isolation Forest and LSTM for anomaly detection.

However, we came across multiple researches citing that “machine identities do not behave like human users that log in at predictable times or exhibit human-like workloads” [11]. This implies that traditional anomaly detection methods are ineffective.

A paper titled, *“Agentic AI for Self-Sovereign Identity: A Decentralized Zero Trust Framework” by Zhu (2025)* [10], while addressing autonomous microservices and Machine-to-Machine authentication, concludes that traditional session-based authentication fails for ephemeral M2M interactions.

Another very crucial research titled, *"Enhancing Trust in Machines Integration with Dirichlet Distribution and Self-Sovereign Identity" by Joshi and Shah (2025)* [10], showcases the closest existing work to our project, NCBA. It proposes a dual-blockchain framework that combines static auth/signed credentials with continuous authentication via ML-enhanced behavioral trust models.

Thus, the drawbacks in the existing behavioral analysis system includes:

- **UEBA models cannot be applied to NPEs** due to the **unique features of Machine Identities** such as Determined communication patterns, Service-to-service communications, Scheduled (cron) tasks.

D. Blockchain Certificate Transparency

The RFC 6962 standardized Certificate Transparency (CT) logs to provide append-only logs of certificate issuances for Identity Management [12]. This help in detecting misissued certificates, but simultaneously are inefficient as they do not record revocation events, usage patterns, or behavioral anomalies.

This issue is also highlighted in the following papers which propose the use of Blockchain as a solution:

"Certificate Transparency Using Blockchain (CTB)" by Manikandan and Sivakumar (2018) showcases the implementation of certificate transparency directly on Hyperledger Fabric. This prevents CAs from issuing certificates without domain owner consent and includes revocation mechanisms. This paper also explicitly notes that "*no existing system specifically detects certificate-based lateral movement",* which also highlights a research gap our project covers [12][13][14]

Another paper titled, *"Anonymous Transactions with Revocation and Auditing in Hyperledger Fabric" by Lu et al. (2017/2021*) at IBM Research CCS, incorporates revocation and auditability into anonymous credential frameworks. We have also taken inspiration from this paper to set up a multi-signature revocation voting (2-of-3) mechanism in our project.

Thus, Blockchain technologies add to Certificate Transparency by providing:

- An immutable and unalterable record
- Decentralized validation

However, the major drawback of current research is:

- This technology alone cannot monitor real-time certificate usage and anomalous activity.

E. Network Intrusion Detection for Certificates

We came across a study by *Kim (2024)* titled, *"A Designing Method of Digital Forensic Snort Application Model"* in the Journal of Digital Forensics [15], which describes packet inspection methodologies through the use of Snort and its adaptation for forensic analysis. It covers various detection methods such as signature, protocol and anomaly-based detection techniques.

Even though this research doesn't center around certificate extraction, it provided us with technical foundations for extending **Snort to handle TLS handshake inspections** that we plan to use for **Certificate Inspection.**

F. Research Gap

From all the above research papers and academic materials around this topic, we have listed the summary of our findings below:

1. SPIFFE-based workload identity solutions reduce credential-related issues, but lack continuous monitoring [1][4].
2. Traditional session-based authentication fails for Ephemeral M2M interactions [9].

3. Current blockchain-based certificate transparency and revocation frameworks built on Hyperledger Fabric, lack integration with behavioral analytics engines [14].
4. The existing systems operate independently without any level of integration [2][10].
5. There is **no comprehensive framework** that incorporates:
    - Temporary certificate handling
    - Machine behavior analysis
    - Detection and monitoring capabilities
    - Automated response mechanisms
    - Unchangeable log management

The current solutions are more reactive in nature and do not protect against lateral movement. Each of the sections we covered above address isolated parts of the **NPE certificate security problem**, but no existing work combines these capabilities into a **unified framework.**

Thus, our project: **NCBA** aims to solve this problem through a cohesive architecture that combines **short-term (non-permanent) certificates**, **behaviour analytics** specific to person (NPP), **real-time monitoring** using **Snort**, **automatic revocation** of **multi-signatures** and a **blockchain-based immutable log** for auditing purposes.

5. Methodology

A. Design Approach

NCBA uses defense-in-depth with three layers:

Preventive Layer: The core idea here is reducing how long a compromised certificate stays valid. Traditional certificates last 6-24 months because manual renewal is painful [7]. NCBA automates renewal so we can drop lifetimes to 1-24 hours without creating an operational nightmare. If an attacker steals a private key, they've got hours to work with instead of months. We trigger renewal at 80% of lifetime expiration to avoid edge cases where clocks drift or renewal fails at the last second.

Detective Layer: In today's system, it will tell you when a certificate expires or violates a static policy, but it doesn't notice when a certificate starts behaving weirdly. NCBA builds behavioral baselines for each certificate instance like what IPs it connects from, what time of day it's active, what services it talks to, what crypto parameters it uses [11][8]. When deviation exceeds thresholds, NCBA flags it.

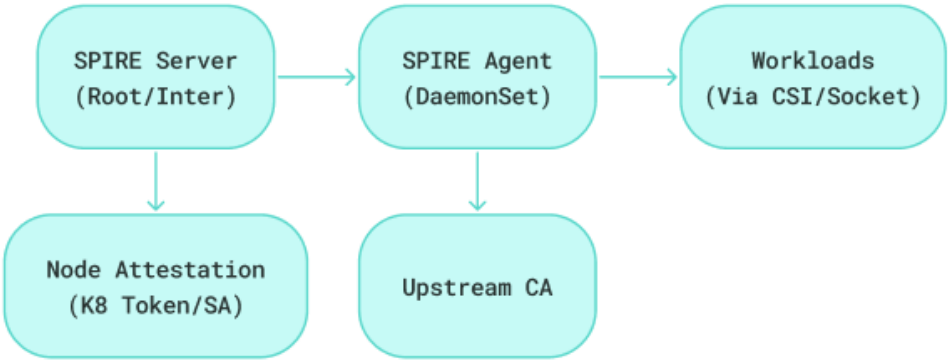
Responsive Layer: This framework automates certificate revocation and network-level blocking. The target is under 30 seconds from anomaly detection to containment, which should be fast enough to disrupt most lateral movement chains [2][6].

The network-centric monitoring approach specifically addresses environments where endpoint agents cannot be installed. Industrial control systems, old OT gear, cheap IoT sensors, etc are devices that often can't handle the overhead of security software. To prevent this overhead, traffic would be analyzed at the network perimeter without touching the endpoints. This aligns with the DoD's stated challenge of securing "devices designed before modern authentication standards existed."

B. Implementation Phases

Phase 1: Baseline Establishment

Certificate Infrastructure: In our project, we will utilize **SPIRE (SPIFFE Runtime Environment)** which is an open source workload identity framework that functions as a specialized **Certificate Authority (CA).** Rather than issuing certificates to humans or static machines, it acts as an automated, short-lived certificate authority for software workloads (containers, microservices, applications, etc.)



Initial Certificate Issuance: We're testing our new credential system on a small mix of apps, databases, and legacy proxies before rolling it out everywhere. These pilot systems get fresh credentials every 24 hours which is much longer than our eventual 15-minute target. It is so that we can iron out configuration issues without constant renewal headaches. Once things run smoothly, we'll tighten the window for better security.

Data Collection: We're pulling network data from two sources, **NetFlow** for traffic patterns and **passive packet capture** for TLS handshakes. Everything lands in a time-series database that keeps raw data for 30 days and aggregated stats much longer. **Network telemetry** collection is a hybrid traffic analysis architecture that functions as a dual-plane data acquisition system.

Service Discovery: We use **Nmap** to scan NPE IP ranges and map what's actually running. It's not perfect as stealth configs and weird ports can hide things, so we cross-check against deployment records and check in with service owners to fill the gaps.

Phase 2: Behavioral Modal Training

Feature Engineering: We look for patterns that are unique to machines and not humans. While people might log in at odd hours from different places, machines have predictable rhythms: cron jobs running on schedule, steady heartbeat checks, consistent encryption settings, and regular talking patterns with other services.

Model Selection: We are using **Isolation Forest** that spots outlier events, while **LSTM autoencoder** catches gradual changes over time. Using both means we don't miss either sudden attacks or slow, sneaky ones.

Training Process: We split our data **80/20** to make sure the model actually learns patterns rather than just memorizing the data. Since we don't have labeled attacks to practice on, we manually check anything that scores high against simulated attack scenarios.

Peer group Analysis: We group similar systems together for example, databases with databases, gateways with gateways. It is so that we don't flag normal database behavior just because it looks different from a web server. This cuts down on false alarms. This contextual normalization technique is called **Cohort-based comparison** that functions as a false positive reduction mechanism [19].

Phase 3: Real-Time Integration

Snort Certificate Inspector (SCI): We built a custom plugin for Snort that digs deep into TLS handshakes to grab certificate serial numbers, cipher details, and server names. It functions as a certificate metadata extraction engine. Regular Snort only skims the surface; this goes deeper to feed our behavioral engine [18].

Risk Scoring: We are using a **threat assessment algorithm that functions** as a multi-factor risk calculator. Unlike binary detection that relies solely on IP reputation, this algorithm acts as a contextual evaluator combining behavioral anomaly (40%), temporal deviation (30%), peer variance (20%), and threat intelligence (10%) into a unified 0-100 risk scale.

Response Rules: The enforcement policy is a tiered intervention framework that functions as an automated response orchestrator. We don't just block or allow. We have four levels: green light (<30), ask for more proof (30-60), slow down and watch (60-85), and shut it down entirely (>85).

Integration APIs: We expose simple REST APIs so components can talk without being tightly coupled. For example endpoints for scoring, checking baselines, and firing alerts all work asynchronously so nothing gets stuck waiting.

Phase 4: Blockchain Integration

Hyperledger Fabric Setup: The distributed ledger is a consortium blockchain network that functions as a tamper-proof audit substrate. We run a private blockchain with three organizations using Raft consensus. It's not a slow public chain, and it's not a regular database that someone could quietly tamper with [20].

Smart Contracts: The on-chain logic is a chaincode trio that functions as a transparent governance mechanism. We have three on-chain programs that handle certificate issuance, log behavioral events, and manage revocations. Revocations need 2-of-3 signatures, so one compromised key can't take everything down.

Integration: We log most things asynchronously so detection stays fast, only waiting for blockchain confirmation when we're actually revoking something critical.

Phase 5: Validation and Testing

Red Team Exercise: We implement adversarial validation which is a controlled penetration test that functions as a detection efficacy benchmark. This exercise acts as a real-world stress test measuring time-to-detection (<5 seconds), evasion resistance, and automated response effectiveness against certificate theft and lateral movement.

Chaos Engineering: We implement resilience verification which is a failure injection protocol that functions as a continuity assurance mechanism. This protocol acts as a fault tolerance validator simulating CA failures, network partitions, Byzantine nodes, and sensor outages to verify graceful degradation.

Load Testing: The performance validation is a traffic simulation suite that functions as a scalability threshold detector. Unlike low-volume testing that misses bottlenecks, this suite acts as a capacity stressor generating 10,000 NPE workloads at 1000-5000 TPS to measure handshake-to-score latency and blockchain commit delays.

False Positive Measurement: The accuracy audit is a manual review process that functions as a precision calibration tool. Unlike automated metrics that accept algorithmic output uncritically, this process acts as a human-in-the-loop validator requiring security analysts to review all CHALLENGE-and-above alerts to maintain <0.1% false positive rates.

6. System Architecture

A. Three Layer Stack Architecture

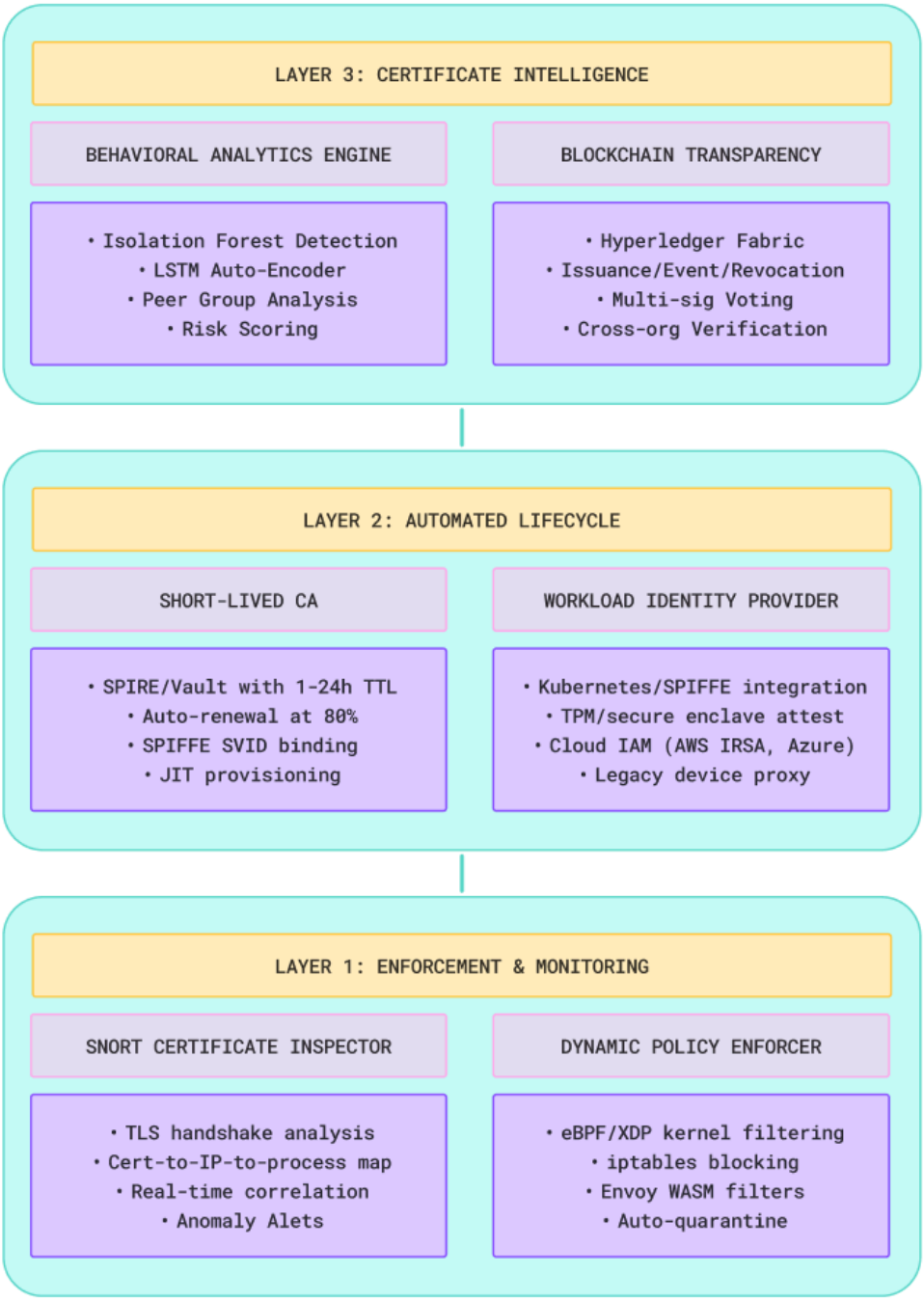
Instead of building one giant, tightly-coupled security system that tries to do everything at once, NCBA splits responsibilities into three distinct layers where each layer has its own job with minimal dependencies on the others.

Layer 1 (The Front Line): Handles the actual blocking and filtering with iptables rules, eBPF programs, and Snort inspection running right on the network.

Layer 2 (The Factory): Manages certificate creation, renewal, and revocation by SPIRE issuing credentials, Vault storing secrets, automated renewal keeping everything fresh.

Layer 3 (The Brain): It runs the analytics like behavioral models spotting anomalies, risk scoring, and decision-making on what to do next.

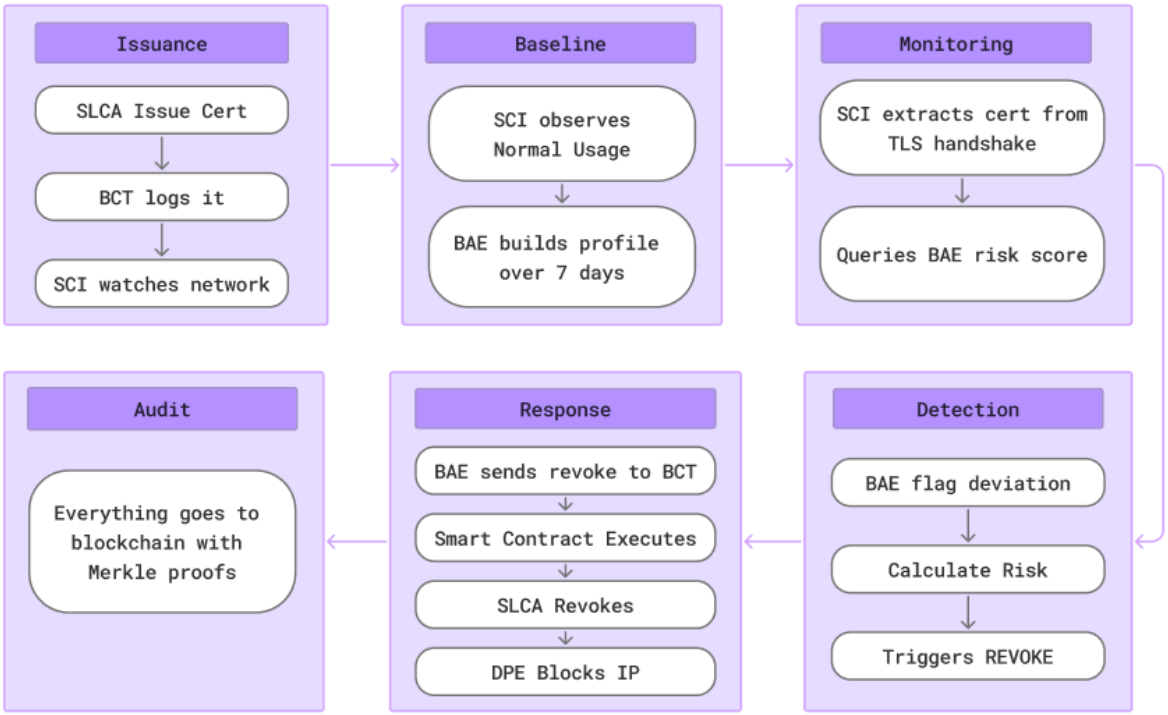
This separation means you can upgrade or fix one layer without breaking the others. If your ML models need retraining, Layer 1 keeps blocking threats. If certificate issuance hiccups, Layer 3 keeps scoring behavior. The system stays up even when individual pieces need attention.



B. Data Flow

In our NCBA operational pipeline, there is a workflow for ongoing monitoring and response. It works as a closed-loop security orchestration system. Unlike reactive incident response workflows, which depend on manual log analysis and ticket-driven fixes after a security breach, this pipeline serves as a real-time process for detecting and containing threats. It automates the steps from issuing certificates to establishing

behavioral baselines, detecting anomalies, revoking cryptographic keys, and logging forensic audits.



C. Tech Stack

- **Backend**: Python (Later Go)
- **Frontend**: React + Typescript
- **Database**: PostgreSQL
- **Time-Series**: InfluxDB
- **Message Queue**: Apache Kafka
- **Cache**: Redis
- **Blockchain**: Hyperledger Fabric
- **CA Infrastructure**: SPIRE
- **IDS**: Snort
- **Container**: Docker + Kubernetes
- **Monitoring**: Prometheus

7. Novel contribution

From our research, most of the existing security tools treat certificate management and threat detection as separate problems. You have one system that issues certificates, another that watches network traffic, and maybe a third that logs events. if they communicate to each other at all, it's through manual scripts or manual ticketing queues. NCBA brings these pieces together into something that actually works as a unified system.

Short-Lived Certificates That Actually Stay Fresh: There are plenty of tools that can issue certificates automatically. But they usually default to 90 days or a year because shorter lifetimes create operational headaches like renewal storms, clock drift issues, failed automation, etc. NCBA pushes this further by defaulting to 24 hours (eventually 15 minutes) and handling the operational complexity through aggressive prefetching, 80% lifetime triggers, and graceful degradation when things go wrong. The novelty isn't just short certificates; it's making them practical at scale without a massive operations team.

Behavioral Models Built for Machines and not Humans: User behavior analytics (UEBA) is for people as it looks for geographic impossibilities, off-hours logins, unusual application access. Machines don't have "off hours" in the human sense. They have cron jobs, health checks, burst scaling, and steady service mesh chatter, hence even more difficult to see any anomalies in their behaviour. NCBA's models are trained specifically on these patterns: inter-arrival times measured in milliseconds, cryptographic consistency (same cipher suite every time), and service dependency graphs that don't change unless something's wrong. We group similar workloads like databases with databases, gateways with gateways. It is so that the normal database behavior doesn't trigger alerts just because it looks different from a web server.

Real-Time Certificate-to-Behavior Correlation: When Snort sees a TLS handshake, most setups check if the certificate is valid and move on. Our NCBA's custom plugin extracts the certificate serial, maps it to a specific workload instance, and immediately checks if that instance is behaving according to its established baseline. This happens in milliseconds, not minutes. If a database certificate suddenly starts talking to a web server, or shows up from a new IP address, the system flags it before the connection fully establishes.

Automated Response Without Human Bottlenecks: Current workflows often generate alerts that sit in a queue waiting for a human to review. NCBA uses tiered automation: low risk gets logged, medium risk triggers additional verification, high risk rate-limits the connection, critical risk revokes the certificate and blocks the IP—all within 30 seconds. Humans review after the fact, not during the incident. The blockchain audit trail ensures this automation is accountable, not opaque.

Network-Centric Protection for Legacy Devices: The Department of Defense explicitly calls out the challenge of securing "devices designed before modern authentication standards existed." Industrial controllers, old OT gear, cheap IoT sensors can't run endpoint agents. NCBA

monitors these at the network perimeter by analyzing their certificate usage, traffic patterns, and cryptographic choices without touching the devices themselves. This extends Zero Trust principles to environments that would otherwise be unprotectable.

What This Adds Up To: NCBA provides the integration: a system where short-lived certificates feed behavioral baselines, where network monitoring triggers automated response, where every action is logged immutably, and where legacy devices get the same protection as cloud-native microservices. It's a practical architecture for a world where machine identities outnumber humans more and more everyday, and where a compromised server certificate can do as much damage as a stolen admin password.

8. Conclusion

The research shows that there is a rapid growth of machine identities in the modern enterprise environment [3][5]. This situation reveals a key flaw in cybersecurity strategy. Current frameworks, which are mainly focused on human access control, struggle when handling non-person entities that show predictable behavior different from human users. The presence of long-lived certificates, often valid for months or even years, creates long periods of vulnerability [6][7]. Sophisticated threat actors can steal the certificate and use that to exploit the network for lateral movement and ongoing access without alerting the security team.

Our project, NCBA tackles this issue with an integrated setup that combines three key innovations: (a) cryptographically bound short-lived certificates that automatically renew every few hours instead of months; (b) machine learning models specifically designed to understand NPE behavior rather than adapting human-focused analytics; and (c) immutable distributed ledger technology that offers tamper-proof audit trails for investigations and regulatory compliance [12][14]. This layered approach cuts the average containment time from months to minutes while allowing for operational effectiveness in limited-resource and legacy environments where adding endpoint controls is not feasible.

The framework's network-focused monitoring strategy tackles the challenge in the Department of Defense Zero Trust guidelines. It seeks to secure older devices that were made before current authentication standards. These devices do not need to have computational power beyond their limits. By connecting cryptographic identity with behavior data at the network edge, NCBA uses Zero Trust principles for the entire machine identity lifecycle.

Future research will examine how to scale to millions of simultaneous NPEs. It will also look at integrating new post-quantum cryptographic standards and addressing autonomous AI-driven workloads that display unpredictable behavior. The current architecture provides a strong foundation for handling machine identity security with the same care usually given to human access. It recognizes that in today's distributed systems, a compromised service certificate poses the same risk to an organization as a compromised privileged user credential.

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